

Institute of Leather Engineering and Technology
University of Dhaka

CHAPTER · Wastewater and Solid Waste Management

IN-PLANT MANAGEMENT FOR POLLUTION REDUCTION

Housekeeping · Segregation · Water · Chemicals · Reuse · Water footprint

A study text in simple English, arranged topic by topic for revision

Contents

Topic	Subject
1	Why in-plant management comes first
2	Housekeeping
3	Segregation of waste streams
4	Reduction of water use
5	Reduction of chemical use
6	Recycling and reuse of chemicals
7	Recycling and reuse of water
8	Water footprint
9	Measuring, benchmarking and making the case
—	Glossary and references

Table 1. The nine topics of this chapter.

1. Why in-plant management comes first

1.1 Two ways to deal with pollution

There are only two ways to deal with the pollution a tannery makes. You can let it form and then treat it at the end of the pipe. Or you can stop it forming in the first place. The whole of the last two chapters described the first way. This chapter describes the second.

DEFINITION

In-plant management means all the measures taken inside the tannery — in the drums, the store and the yard — that reduce the amount of water, chemical and pollution leaving the process, before any treatment plant is reached.

It is also called **cleaner production**, or pollution prevention at source.

1.2 The order of preference

Environmental engineering follows a fixed order of preference. Each step is cheaper and more certain than the one below it.

Avoid → Reduce → Reuse and recycle → Treat → Dispose

End-of-pipe treatment sits near the bottom. It is necessary, but it is the most expensive answer and the least reliable, because it depends on a plant running well every day. Everything above it in the list works by simply not creating the problem.

KEY IDEA

A chemical never bought is cheaper than one recovered, which is cheaper than one treated. Every kilogram that does not fix on the hide is paid for twice — once to the supplier, and once to the treatment plant.

1.3 Why it matters most in a tannery

A tannery is a wet process with a poor fixation rate. About **500 kg of chemicals are used for every tonne of raw hide**, and **less than 20% of them stay in the finished leather**. Only about half the collagen in the hide ends up as leather. Everything else leaves in the effluent or the solid waste, and the treatment plant must then deal with it.

In-plant management attacks the problem where it is smallest and cheapest — in a drum, not in a clarifier.

1.4 The size of the problem, in numbers

These are the loads a conventional tannery sends to treatment for every tonne of raw hide it processes. Learn the bovine column; the others show how much worse sheepskin processing is.

Parameter	Bovine hides	Pig skins	Sheepskins
COD	145 – 230 kg/t	140 – 320 kg/t	330 – 1,005 kg/t
BOD₅	48 – 86 kg/t	52 – 115 kg/t	135 – 397 kg/t
Suspended solids	85 – 155 kg/t	70 – 135 kg/t	175 – 352 kg/t
Total dissolved solids	300 – 520 kg/t	180 – 500 kg/t	varies widely
Chlorides	145 – 220 kg/t	80 – 240 kg/t	210 – 640 kg/t
Chromium (III)	3 – 7 kg/t	3 – 6 kg/t	9 – 15 kg/t
Sulphide	2 – 9 kg/t	3 – 7 kg/t	6 – 20 kg/t

Table 2. Pollution load per tonne of raw hide, conventional processing (World Bank EHS Guidelines).

Read the chromium row against Topic 6. A tannery sends 3–7 kg of chromium per tonne of hide to the drain. Recovering it is worth money, and leaving it there makes every tonne of sludge a hazardous waste.

KEY POINTS — learn these

1. In-plant management stops pollution being made; end-of-pipe treatment removes it after it is made.
2. The order of preference is avoid, reduce, reuse and recycle, treat, dispose.
3. It is also called cleaner production or pollution prevention at source.
4. About 500 kg of chemicals are used per tonne of raw hide, and under 20% stays in the leather.
5. It is the cheapest option because unused chemicals are paid for twice — to the supplier and to the treatment plant.

REMEMBER THESE NUMBERS

Chemical use $\approx$ **500 kg per tonne** of raw hide; **under 20%** is retained in the leather.
 Conventional bovine load per tonne: COD **145–230 kg**, BOD₅ **48–86 kg**, SS **85–155 kg**, TDS **300–520 kg**, chromium **3–7 kg**, sulphide **2–9 kg**.

LIKELY EXAM QUESTIONS

1. Define in-plant management and distinguish it from end-of-pipe treatment.
2. State the order of preference in pollution control and explain why treatment sits near the bottom.
3. About how many kilograms of chemicals are used per tonne of raw hide, and what share stays in the leather?
4. Quote the conventional bovine COD and chromium loads per tonne, and explain what each implies for the treatment plant.

2. Housekeeping

Housekeeping means the simple, daily discipline of running the plant properly. These are the cheapest measures in the whole course. None of them needs capital equipment. All of them need supervision. Together they are routinely worth **10–25% of the water and chemical load**.

Measure	What it means in practice
Measure the float	Drums are run to a marked ratio, not to a hose left open
Stop the running hose	Trigger nozzles and timed valves. An open 25 mm hose wastes about 25 m ³ in a shift
Contain the spills	Drip trays under drums, kerbed chemical stores, bunded pits
Fix the leaks	Worn drum seals, dripping valves and overflowing pits are all unmetered load
Meter and record	A meter for each section and a daily logbook. Nothing that is not measured is ever reduced
Train and supervise	The same drum and the same recipe, run by two operators, can use twice the water

Table 3. The six housekeeping measures.

KEY IDEA

Spilt chemical is bought twice — once from the supplier, and once from the CETP. Housekeeping costs nothing but attention, and it is the first thing an auditor looks at.

KEY POINTS — learn these

1. Housekeeping needs no capital, only supervision, and saves 10–25% of water and chemical load.
2. The six measures: measure the float, stop the running hose, contain spills, fix leaks, meter and record, train and supervise.
3. Metering is the foundation. What is not measured cannot be managed or reduced.
4. Operator variation is a real source of waste: the same recipe run by different people uses different amounts.

REMEMBER THESE NUMBERS

Housekeeping is worth **10–25%** of water and chemical load.

An open **25 mm hose** wastes about **25 m³ in one shift**.

LIKELY EXAM QUESTIONS

1. Name the six housekeeping measures and explain what each achieves.
2. Why is metering described as the foundation of in-plant management?
3. "Spilt chemical is bought twice." Explain this statement.

3. Segregation of waste streams

Four streams in a tannery must not meet before treatment. Mixing them destroys the chance of recovering any of them, and the mixture is harder to treat than any of its parts.

Stream	Where it comes from	If kept separate	If mixed
Chrome liquor	Spent tanning and wet-back floats	Precipitated and returned to the tanyard	Chrome enters all the sludge, which becomes hazardous waste
Sulphide liquor	Lime and unhairing floats	Hair screened out, then catalytic oxidation	Meets acid and forms H_2S ; kills the biological stage
Salt-laden liquor	Curing yard drainage, spent pickle	Held for evaporation or reuse	10,000–40,000 mg/L TDS passes through the whole plant
Clean water	Cooling, boiler blowdown, roof runoff	Reused directly, or discharged	Dilutes the effluent and makes the plant larger than it needs to be

Table 4. The four streams that must be kept apart.

Note the last row. Clean water is not a pollutant, but adding it to the effluent is still a fault. It increases the volume that must be pumped, treated and paid for, without adding any pollutant load.

KEY IDEA

Segregation costs pipework and discipline. Mixing costs a hazardous classification on every tonne of sludge the plant produces. Every recycling method in Topic 6 depends absolutely on segregation — none of those liquors survives being mixed.

3.1 Treating the sulphide stream at source

The sulphide stream is the one that must be dealt with immediately, because it is dangerous. Spent lime liquor holds **1,500–3,000 mg/L of sulphide**. If it is allowed to mix with the acid streams, hydrogen sulphide gas is released in the sewer. This has killed sewer and tannery workers.

The standard answer is **catalytic oxidation**. The segregated liquor is screened to remove hair, then aerated in a tank. Air oxidises the sulphide to thiosulphate and, in smaller part, to sulphate. A manganese salt (manganese sulphate) is dosed as a catalyst to speed the reaction.

Item	Value
Sulphide in spent lime liquor	1,500 – 3,000 mg/L (200–600 mg/L once diluted by other streams)
Catalyst	Manganese sulphate, dosed in small quantity
Oxygen required	About 2 kg of oxygen per kg of sulphide, with excess air supplied
Aeration time	About 8 hours for complete oxidation
Target after oxidation	Below 1 mg/L as S^{2-} (ECR 2023 limit is 2.0 mg/L)

Table 5. Catalytic sulphide oxidation — operating conditions.

KEY POINTS — learn these

1. Four streams must be kept separate: chrome liquor, sulphide liquor, salt-laden liquor, and clean water.
2. Mixing destroys the option to recover anything, and the mixture is harder to treat than its parts.
3. Sulphide meeting acid produces hydrogen sulphide — a danger to workers and fatal to the biological stage.
4. Adding clean water to the effluent is a fault: it increases volume and cost without adding load.
5. Segregation is the precondition for all chemical recycling.

REMEMBER THESE NUMBERS

Salt-laden liquor carries **10,000–40,000 mg/L TDS**.

Spent lime liquor holds **1,500–3,000 mg/L sulphide**; diluted in the mixed stream, **200–600 mg/L**.

Catalytic oxidation: **manganese sulphate** catalyst, **≈ 2 kg O₂ per kg S²⁻**, **8 hours** aeration, target **below 1 mg/L**.

LIKELY EXAM QUESTIONS

1. Name the four streams that must be segregated, and state the consequence of mixing each.
2. Why is it a fault to send cooling water to the effluent, even though it is clean?
3. Describe catalytic sulphide oxidation: the catalyst, the oxygen requirement, the aeration time and the target concentration.
4. Explain why segregation is the precondition for chrome recovery.

4. Reduction of water use

A conventional tannery uses **30–40 m³ of water per tonne of raw hide**. Good practice reaches **12–18 m³/t**. The difference is not one clever technology. It is a series of decisions about floats and washes.

4.1 What the benchmarks say

Two published benchmarks are worth knowing. The World Bank EHS Guidelines give the range found in practice across the world. The European BAT reference document gives the level a well-run plant should reach.

Source and basis	Bovine hides	Other raw stock
World Bank EHS — range in practice	12 – 50	Pig 32–69; sheep 110–265
EU BAT — raw to wet blue, salted hides	13 – 18	unsalted 10–15
EU BAT — post-tanning	6 – 10	—
EU BAT — total, salted hides	19 – 28	unsalted 16–25
UNIDO model tannery target	not more than 25	—

Table 6. Water consumption benchmarks, m³ per tonne of raw hide.

Note the two BAT lines. Most of the water is used before the leather is even tanned. If you want to save water, the beamhouse is where to look first.

4.2 The float, and what a short float means

DEFINITION

The **float** is the volume of water in a drum, given as a percentage of the pelt or hide weight. A 100% float on 1,000 kg of pelt means 1,000 litres of water.

A conventional tanning or post-tanning float runs at 150–200%. A short float runs at 40–80%. Less water means the chemicals are more concentrated, so they penetrate and fix better. A short float therefore saves water **and** improves exhaustion at the same time. This is unusual: most environmental measures cost something.

4.3 The measures

Measure	How it works	Typical saving
Short floats	Tanning and post-tanning run at 40–80% float instead of 150–200%	15–30% of process water
Batch washing	Fill, run, drain — instead of a hose running while the drum turns	Up to 50% of wash water
Fewer, better washes	Two efficient washes beat five careless ones; check by drain conductivity	10–20%
Reuse of clean streams	Cooling water, final rinses and RO permeate returned upstream	10–25%
Automatic drum controls	Timed, metered filling, so volume no longer depends on the operator	5–15%

Table 7. Measures that reduce water use.

Every cubic metre not used is a cubic metre not pumped, not heated, not treated and not discharged. The saving counts four times over.

WORKED EXAMPLE — water balance and savings

A tannery processes **6 t of raw hide per day at 33 m³/t** and discharges to the estate CETP at **Tk 45 per m³**. Short floats, batch washing and cooling-water reuse are introduced, bringing use down to 20 m³/t.

Present water use. $6 \times 33 = 198 \text{ m}^3/\text{d}$

After the measures. $6 \times 20 = 120 \text{ m}^3/\text{d}$

Water saved. $198 - 120 = 78 \text{ m}^3/\text{d}$, a saving of **39%**

CETP charge avoided. $78 \times 45 = \text{Tk } 3,510 \text{ per day}$, or about **Tk 1.05 million per year** over 300 working days

COD load before. $198 \text{ m}^3 \times 2,500 \text{ mg/L} = 495 \text{ kg/d}$

COD load after. $120 \text{ m}^3 \times 3,600 \text{ mg/L} = 432 \text{ kg/d}$

Comment. Watch the concentration. The load falls by only 13%, but the effluent strengthens from 2,500 to 3,600 mg/L. A stronger, smaller flow is welcome to a biological stage, but it punishes any unit that was sized on concentration rather than on load.

KEY POINTS — learn these

1. Conventional use is 30–40 m³/t of raw hide; good practice is 12–18 m³/t.
2. The five measures: short floats, batch washing, fewer and better washes, reuse of clean streams, automatic drum controls.
3. Every cubic metre saved is saved four times: pumping, heating, treatment and discharge.
4. Saving water raises the concentration of the effluent even though it lowers the total load.

REMEMBER THESE NUMBERS

Water use: conventional **30–40 m³/t**, good practice **12–18 m³/t**.

EU BAT for salted bovine hides: raw to wet blue **13–18**, post-tanning **6–10**, total **19–28 m³/t**.

World Bank range: bovine **12–50**, pig **32–69**, sheepskin **110–265 m³/t**.

Short floats: **40–80%** instead of **150–200%**.

Savings: short floats **15–30%**, batch washing **up to 50%**, better washes **10–20%**, reuse **10–25%**, automatic controls **5–15%**.

LIKELY EXAM QUESTIONS

1. Define the float and explain why a short float saves water and improves exhaustion at the same time.
2. Quote the EU BAT water consumption levels for salted bovine hides, split between the beamhouse and post-tanning. Which stage uses more?
3. Name five measures that reduce water use and give the saving from each.
4. A tannery cuts water use by 40%. Explain what happens to the effluent concentration, and why that matters to the treatment plant.

5. Reduction of chemical use

Every kilogram of chemical that does not fix on the hide leaves in the effluent. There are only two levers: **offer less**, and **fix more of what is offered**.

Measure	What it does
High-exhaustion chrome	Controlled pH, temperature and basification raise exhaustion from 60–70% to 90–95%
Hair-save unhairing	Hair is screened out rather than dissolved, cutting beamhouse COD by 30–50%
Low-salt curing	Chilling, short-term preservation or biocides replace 40% salt on hide weight
CO₂ deliming	Replaces ammonium salts, removing ammonia nitrogen at source
Enzyme assistance	In soaking, unhairing and bating — a lower chemical offer and shorter processes
Reduced sulphide	Amine-assisted or enzyme-assisted systems, where hide quality allows

Table 8. Measures that reduce chemical use.

5.1 Chrome: the offer and the exhaustion

Chrome is worth studying closely, because it is the most valuable chemical in the tannery and the most damaging one in the effluent.

Item	Value
Typical offer	1.5 – 2.5% Cr ₂ O ₃ on pelt weight; about 2.0% conventionally
Optimised offer	as low as 1.3% Cr ₂ O ₃ on pelt weight
Conventional exhaustion	60 – 80% of the offer taken up by the hide
High-exhaustion systems	up to 95 – 98% uptake
Chromium reaching the effluent	3 – 7 kg per tonne of raw hide, conventionally

Table 9. Chrome offer and exhaustion.

High exhaustion is achieved by controlling three things: the pH at the end of tanning (basification), the temperature, and the time allowed. Masked and self-basifying chrome products help. The result is that far less chromium reaches the drain — and, importantly, far less is left to recover, which changes the economics of Topic 6.

Low-salt curing attacks the TDS problem at its only true source. As the tertiary treatment chapter showed, once salt is dissolved only reverse osmosis will remove it, at great cost. Conventional curing uses about 40% salt on hide weight, and every kilogram of it ends up in the effluent.

KEY IDEA

A chemical never bought is cheaper than one recovered, which is cheaper than one treated. This single sentence ranks every measure in this chapter.

KEY POINTS — learn these

1. Only two levers exist: offer less chemical, and fix more of what is offered.
2. High-exhaustion chrome raises exhaustion from 60–70% to 90–95%.
3. Hair-save unhairing screens the hair instead of dissolving it, cutting beamhouse COD by 30–50%.
4. Low-salt curing is the only real answer to TDS, because dissolved salt can only be removed by RO.
5. CO₂ deliming removes ammonia nitrogen at source by replacing ammonium salts.

REMEMBER THESE NUMBERS

Chrome offer **1.5–2.5% Cr₂O₃ on pelt weight**, typically 2.0%; optimised to **1.3%**.

Chrome exhaustion: conventional **60–80%**, high-exhaustion **up to 95–98%**.

Hair-save unhairing cuts beamhouse COD by **30–50%**; enzymatic dehairing up to **40–50%**.

Conventional curing uses about **40% salt on hide weight**.

LIKELY EXAM QUESTIONS

1. State the two levers available for reducing chemical use, and give an example of each.
2. Explain how high-exhaustion chrome tanning works and what it achieves.
3. Why is low-salt curing the only real answer to the TDS problem?
4. What does CO₂ deliming replace, and what pollutant does it remove at source?

6. Recycling and reuse of chemicals

Four tannery liquors can genuinely be recycled at plant scale. Each one depends completely on segregation — none of them survives being mixed with another stream.

Liquor	Treatment	Returned as	The condition
Spent chrome	Precipitate with MgO or NaOH, settle, then redissolve in H ₂ SO ₄	20–30% of the fresh chrome offer	Tanyard stream kept separate
Spent lime–sulphide	Screen out the hair, settle, top up sulphide and lime	30–50% of the unhairing float	Solids removed every cycle
Spent pickle	Filter, then adjust salt and acid back to strength	Most of the pickle float	Protein build-up limits the number of cycles
Spent vegetable liquor	Counter-current pit system	Progressive reuse of the tannin	Long-standing practice, not new

Table 10. The four recyclable liquors.

Chrome recovery is the most important of the four. It pays twice: it returns part of the chrome offer to the tanyard as a raw material, and it takes the chromium out of the sludge. Sludge without chromium is not hazardous waste, and its disposal cost falls sharply.

6.1 How chrome recovery is actually done

The process is simple chemistry, carried out in two tanks beside the tanyard.

Step	What is done
1 Collect	Spent chrome float from the segregated tanyard stream, screened of solids
2 Precipitate	Add magnesium oxide (MgO), or sodium hydroxide or carbonate, to raise the pH. The optimum is pH 8.5 – 9.0 . Chromium comes out as chromium hydroxide, Cr(OH) ₃
3 Settle	Let the sludge stand overnight, or up to 24 hours . Decant or filter the clear liquid off
4 Redissolve	Dissolve the filter cake in concentrated sulphuric acid, about 2 kg H₂SO₄ per kg of Cr₂O₃ , down to pH 2.5
5 Return	Add the recovered basic chromium sulphate back to the tanning float, replacing part of the fresh offer

Table 11. Chrome recovery — operating conditions.

MgO is usually preferred to caustic soda. It works over a wider pH band and the sludge redissolves less readily, so the process is more forgiving of an untrained operator.

KEY IDEA

There is a limit, and an irony in it. Recovered chrome should replace **at most 30% of the fresh offer**, or leather quality suffers. And the better your exhaustion, the less there is to recover: a conventional float yields 25–30% of the offer, but a high-exhaustion float yields only 1–5%. High exhaustion and chrome recovery are alternatives, not partners.

KEY POINTS — learn these

1. Four liquors are recyclable: spent chrome, spent lime–sulphide, spent pickle, spent vegetable liquor.
2. Every one of them depends on segregation. A mixed liquor cannot be recycled.
3. Chrome recovery: precipitate with MgO at pH 8.5–9.0, settle 24 h, redissolve in H₂SO₄ to pH 2.5, return to the float.
4. Chrome recovery pays twice — recovered chrome, and sludge that is no longer hazardous.
5. Recovered chrome should replace at most 30% of the fresh offer.
6. High exhaustion and chrome recovery are alternatives: the better the exhaustion, the less there is to recover.
7. Spent pickle recycling is limited by protein building up over successive cycles.

REMEMBER THESE NUMBERS

Precipitation at **pH 8.5–9.0**; settle **overnight to 24 h**; redissolve with about **2 kg H₂SO₄ per kg Cr₂O₃** to **pH 2.5**.

Recovery from a conventional float **25–30%** of the offer; from a high-exhaustion float only **1–5%**.

Recovered chrome may replace at most **30%** of the fresh offer.

Lime–sulphide recycling returns **30–50%** of the unhairing float.

LIKELY EXAM QUESTIONS

1. Name the four recyclable tannery liquors and state what each is returned as.
2. Describe the five steps of chrome recovery, giving the pH at each stage and the chemicals used.
3. Why is magnesium oxide usually preferred to caustic soda as the precipitant?
4. Explain why high-exhaustion tanning and chrome recovery are alternatives rather than partners.
5. What limits the number of times a pickle liquor can be reused?

7. Recycling and reuse of water

Water does not need to be clean to be useful. It only needs to be clean enough for the next job. The rule is to **cascade**: send the cleanest water to the most sensitive step, then pass it down to less sensitive ones.

Stream	Where it can go
Cooling and boiler water	Never touched the hide — reuse it directly, or send it to soaking
Final rinses	Lightly loaded — return them as the first wash of the next batch
Soak and wash liquors	Screened and settled — back to the curing yard or the first soak
RO permeate	300–400 mg/L TDS — the only stream clean enough to return to a tanning float
Treated CETP effluent	Floor washing, dust suppression, gardening — never into a float

Table 12. Water streams and where each can be reused.

KEY IDEA

The hierarchy is set by **salt, not by BOD**. Anything carrying dissolved solids cannot go back into a float, however clear it looks. This is why RO permeate is the only treated stream fit to return to the process.

KEY POINTS — learn these

1. The principle is cascading: cleanest water to the most sensitive step, then downward.
2. Cooling and boiler water never touched the hide and can be reused directly.
3. Final rinses become the first wash of the next batch.
4. RO permeate at 300–400 mg/L TDS is the only treated stream fit for a tanning float.
5. Treated CETP effluent may be used for washing floors and dust suppression, never in a float.
6. The limit on reuse is TDS, not organic load.

REMEMBER THESE NUMBERS

RO permeate is **300–400 mg/L TDS**.

Reuse of clean streams saves **10–25%** of total water use.

LIKELY EXAM QUESTIONS

1. Explain the cascade principle in water reuse.
2. Name five streams available for reuse in a tannery and say where each can go.
3. Why is the reuse hierarchy set by salt rather than by BOD?

8. Water footprint

The final idea in this chapter is a way of measuring the whole water burden of a product, not just the water passing through the factory meter.

DEFINITION

The **water footprint** (ISO 14046) is the total volume of fresh water used to make a product, counted over its whole life cycle, and separated into three colours.

Component	What it counts
Blue	Surface or ground water taken out and not returned to the same catchment — the tannery's own 12–40 m ³ /t
Green	Rainwater consumed in growing the feed for the animal — dominant for any livestock product
Grey	The clean water that would be needed to dilute the discharge back to standard — this is where treatment appears

Table 13. The three components of a water footprint.

For leather the total is about **17,000 litres per kilogram of bovine leather**, cradle-to-gate. The tannery's own abstraction is only a few percent of that. Almost all of it is green water, upstream, in raising the animal — a component the tannery does not control.

KEY IDEA

Blue water alone flatters the sector, because it hides the enormous green component upstream. **Grey water is the part that in-plant management actually moves** — every kilogram of pollutant not discharged is dilution water not needed.

KEY POINTS — learn these

1. The water footprint is defined by ISO 14046 and counts the water behind the product, not the water through the meter.
2. Blue water is abstracted and not returned; green water is rainwater used to grow feed; grey water is the water needed to dilute the discharge to standard.
3. Bovine leather is about 17,000 litres per kilogram, cradle-to-gate.
4. The footprint is dominated by green water upstream, which the tannery does not control.
5. In-plant management works mainly on the grey component, by reducing what is discharged.

REMEMBER THESE NUMBERS

Standard: **ISO 14046**.

Bovine leather footprint $\approx$ **17,000 litres per kg**, cradle-to-gate.

The tannery's own blue water is **12–40 m³ per tonne** of raw hide.

LIKELY EXAM QUESTIONS

1. Define the water footprint and name its three components.
2. Which component dominates the footprint of leather, and why can the tannery not control it?
3. Which component does in-plant management actually reduce, and how?
4. Why is it misleading to report blue water alone for the leather sector?

9. Measuring, benchmarking and making the case

Nothing in this chapter can be managed until it is measured, and no manager will pay for it until the case is made in money. This last topic is about both.

9.1 The water and material balance

A **water balance** accounts for every cubic metre entering and leaving the tannery. A **material balance** does the same for a chemical. Both are built the same way.

- Fit a meter on the mains supply, and one on each section — beamhouse, tanyard, post-tanning, finishing, boiler.
- Record the reading at the same time every day, in a logbook that a named person signs.
- Record the production over the same period, in tonnes of raw hide.

- Divide one by the other. That is your specific consumption, in m³ per tonne.
- Compare it with the benchmark in Topic 4. The gap between your figure and the benchmark is your opportunity, in numbers.

A balance almost never closes on the first attempt. The difference between what you bought and what you can account for is itself a finding: it is leaks, overflows and unmetered hoses.

9.2 The indicators to track

Indicator	Unit	Good practice
Specific water consumption	m ³ per tonne raw hide	12 – 18; BAT total 19–28 for salted hides
Specific chemical consumption	kg per tonne raw hide	below the 500 kg/t conventional figure
Specific COD load	kg COD per tonne raw hide	below the 145–230 kg/t conventional range
Chrome exhaustion	% of offer taken up	90% or better
Effluent volume	m ³ per tonne raw hide	close to water consumption; a large gap means dilution
Sludge produced	kg dry solids per tonne	falls when chrome is recovered and floats are shortened

Table 14. Key performance indicators for in-plant management.

KEY IDEA

Always express the indicator **per tonne of raw hide**, never as a monthly total. A monthly total falls when production falls, which tells you nothing. Specific consumption tells you whether the plant is being run better.

9.3 What the effluent must reach

It is worth seeing what a well-run plant is expected to achieve at the outfall, because in-plant management is what makes those numbers reachable at sensible cost.

Parameter	EU BAT-AEL	Bangladesh ECR 2023
COD	200 – 500 mg/L	200 mg/L
BOD ₅	15 – 25 mg/L	50 mg/L
Suspended solids	below 35 mg/L	150 mg/L
Ammoniacal nitrogen	below 10 mg/L	—
Total chromium	0.3 – 1 mg/L	2.0 mg/L
Sulphide	below 1 mg/L	2.0 mg/L

Table 15. EU BAT-associated emission levels for direct discharge, compared with ECR 2023.

Bangladesh limits are less strict than the European ones on solids and BOD, and about the same on COD. A tannery that reaches BAT levels will comply anywhere.

9.4 Making the financial case

A manager will approve a measure if the payback is short. The calculation is simple.

$$\text{Payback period (years)} = \text{capital cost} \div \text{annual saving}$$

WORKED EXAMPLE — payback on a chrome recovery unit

A tannery processes **6 t of raw hide per day, 300 days a year**. Chrome offer is 2.0% Cr_2O_3 on pelt weight, and chromium reaching the drain is **5 kg per tonne of raw hide**. Basic chromium sulphate costs **Tk 90 per kg**. A recovery unit costs **Tk 1,800,000** and its chemicals and power cost **Tk 250,000 a year**.

Step 1 — Chromium going to drain. $6 \times 5 = 30 \text{ kg/d}$, or $30 \times 300 = 9,000 \text{ kg per year}$

Step 2 — What can be recovered. Take 27% of the offer as recoverable and allow that recovered chrome may replace only 30% of the fresh offer. Recovered and usable $\approx 6,000 \text{ kg per year}$

Step 3 — Value of the chrome. $6,000 \times 90 = \text{Tk } 540,000 \text{ per year}$

Step 4 — Sludge saving. Removing chromium takes the sludge out of the hazardous class. Assume disposal falls from Tk 6,000 to Tk 1,500 per tonne on 200 t/yr: saving **Tk 900,000 per year**

Step 5 — Net annual saving. $540,000 + 900,000 - 250,000 = \text{Tk } 1,190,000 \text{ per year}$

Step 6 — Payback. $1,800,000 \div 1,190,000 = 1.5 \text{ years}$

Comment. Note where the money actually came from. The recovered chrome is worth Tk 540,000, but the avoided hazardous-waste charge is worth almost twice that. Chrome recovery is usually justified by the sludge, not by the chrome. The figures used here are illustrative — use your own plant's prices.

9.5 Why it is not always done

If in-plant management is so cheap and pays back so quickly, why do so many tanneries not do it? The barriers are real and are worth naming.

- **No measurement.** Without meters, nobody knows what is being used, so no saving can be proved.
- **Split incentives.** The production manager is judged on output, not on water. The cost appears in someone else's budget.
- **Fear for quality.** Short floats and reduced chemical offers are believed to risk the leather. The fear is reasonable and is answered only by trials on the plant's own hides.
- **No capital, and no access to it.** A small tannery may not be able to borrow Tk 1.8 million however good the payback.
- **Charging by production, not by flow.** If the CETP bills by hides processed, saving water saves nothing.
- **Skills and supervision.** Every measure in this chapter needs someone who understands it and is accountable for it.

The first and the last of these are the ones a young engineer can change without any capital at all.

KEY POINTS — learn these

1. Nothing is managed until it is measured. Fit meters by section and keep a signed daily logbook.
2. Always express consumption per tonne of raw hide, not as a monthly total.
3. A balance that does not close is itself a finding: the gap is leaks and unmetered use.
4. Track water, chemical and COD per tonne, chrome exhaustion, effluent volume and sludge.
5. $\text{Payback} = \text{capital cost} \div \text{annual saving}$. Chrome recovery is usually justified by the avoided sludge cost, not by the chrome.
6. The main barriers are no measurement, split incentives, fear for quality, lack of capital, charging by production, and lack of supervision.

REMEMBER THESE NUMBERS

EU BAT-AEL: COD **200–500**, BOD₅ **15–25**, SS **below 35**, total chromium **0.3–1**, sulphide **below 1 mg/L**.

$\text{Payback} = \text{capital cost} \div \text{annual saving}$.

LIKELY EXAM QUESTIONS

1. Describe how you would carry out a water balance in a tannery, and say what a balance that fails to close tells you.
2. Why must consumption always be expressed per tonne of raw hide?
3. List six key performance indicators for in-plant management and give the good-practice value for each.
4. A recovery unit costs Tk 1.8 million and saves Tk 1.19 million a year. Calculate the payback period and comment on whether it should be approved.
5. Give four reasons why tanneries fail to adopt cleaner production, and say which of them an engineer can change without capital.

Think about it — a decision to discuss

THE SITUATION

Your tannery pays **Tk 45 per m³** at the CETP and uses **33 m³ per tonne of hide**. Your managing director will fund exactly **one** of the following this year.

1. Automatic float metering on every drum, and a written washing procedure
2. A chrome recovery unit on the segregated tanyard stream
3. Hair-save unhairing in the beamhouse
4. A hose, nozzle and logbook discipline programme, with a supervisor accountable for it

Which one pays back first? Which one reduces the CETP's problem most? Are they the same one?

Two of these cost almost nothing. That is not the same as being worth the least.

Glossary and references

Term	Meaning
Basification	Raising the pH at the end of chrome tanning so that the chrome fixes on the hide
BAT / BAT-AEL	Best Available Techniques, and the emission levels associated with them (EU reference document)
Cascade reuse	Passing water from a sensitive step down to progressively less sensitive ones
Cleaner production	Another name for in-plant management: preventing pollution at source
End-of-pipe treatment	Removing pollution after it has been made, in a treatment plant
Catalytic oxidation	Aerating segregated lime liquor, with a manganese salt as catalyst, to destroy sulphide
Exhaustion	The percentage of a chemical offered that actually fixes on the hide
Float	The volume of water in a drum, expressed as a percentage of pelt or hide weight
Hair-save unhairing	Unhairing in which the hair is screened out whole instead of being dissolved
Housekeeping	The daily discipline of running the plant properly — metering, containing, fixing, training
ISO 14046	The international standard for water footprint assessment
Offer	The quantity of a chemical added, as a percentage of hide or pelt weight
RO permeate	The clean water produced by reverse osmosis, typically 300–400 mg/L TDS
Payback period	Capital cost divided by annual saving — the years taken to repay an investment
Segregation	Keeping waste streams separate so that each can be recovered or treated properly
Specific consumption	Water or chemical used per tonne of raw hide — the only fair basis for comparison
TDS	Total dissolved solids, in mg/L — dissolved salt, removable only by membranes
Water footprint	The total fresh water behind a product, split into blue, green and grey

Table 16. Terms used in this chapter.

References

- World Bank / IFC — *Environmental, Health and Safety Guidelines for Tanning and Leather Finishing*. Source of the pollution load and water consumption ranges in Topics 1 and 4.
- European Commission — *Best Available Techniques (BAT) Reference Document for the Tanning of Hides and Skins*, and Commission Implementing Decision 2013/84/EU. Source of the BAT water levels and BAT-AELs.
- UNIDO — *Chrome Management in the Tanyard*. Source of the chrome offer, exhaustion and recovery figures in Topics 5 and 6.
- Buljan, J. and Král, I. — *Benchmarking in the Tanning Industry* (UNIDO).
- UNIDO — *Introduction to Treatment of Tannery Effluents*, and guidance on chrome recovery and cleaner production.
- Scholz, W. — *Key Technologies for Tannery Wastewater Treatment* (IULTCS).
- Buljan, J. and Král, I. — *The Framework for Sustainable Leather Manufacture* (UNIDO).
- ISO 14046 — *Environmental management: Water footprint. Principles, requirements and guidelines*.

